

proof steps of Lagrangian Multiplier Theorem.

Main idea:

connect what know from unconstrained problems with the constrained ones.

Step 1

consider a sequence of functions.

$$\Rightarrow \underline{f^k(x)} = f(x) + \underline{k \cdot |h(x)|^2} + \frac{\alpha}{2} \|x - x^*\|^2,$$

$k=1, 2, 3, \dots$ x^* is the local min. within a radius of δ

let's consider the follow problem

$$\min f^k(x) \quad \text{s.t.} \quad \|x - x^*\| \leq \delta \quad (2)$$

call x^k to be global min for problem (2)

Will show:

when $k \rightarrow \infty$

mean x^k close to x^*

Assume
(a) (b) True
for now

(a)
(b)

x^k becomes an unconstrained min of (2)

$x^k \rightarrow x^*$

suppose (a) is true x^k is an unconstrained min of (2)

Step 2 $\nabla f^k(x^k) = 0 \Rightarrow \nabla f(x^k) + 2k \nabla h(x^k) h(x^k) + \alpha (x^k - x^*) = 0$

Recall:

$$\nabla f(x^*) + \lambda^* \nabla h(x^*) = 0 \quad \text{of } (x^*) + \left(\lim_{k \rightarrow \infty} 2k \cdot h(x^k) \right) \cdot \nabla h(x^*) = 0$$

$$h(x) \in \mathbb{R}^{m \times 1}$$

$$\nabla h(x) \in \mathbb{R}^{n \times m}$$

$$x \in \mathbb{R}^n$$

m # constant

$\nabla h(x^*)$ has full column rank.

$$\Rightarrow \nabla h(x^*)^T \cdot \nabla h(x^*) \text{ invertible}$$

$$m \times m$$

#variable
regularity

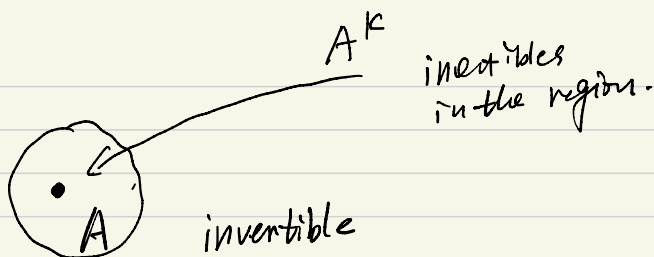
$$\nabla h(x^*)$$

Combined with the fact that

$$x^k \rightarrow x^*$$

$$\Rightarrow$$

$$\nabla h(x^k)^T \nabla h(x^k) \text{ invertible}$$



Step 3

$$\nabla f(x^k) + 2k \nabla h(x^k) h(x^k) + 2(x^k - x^*) = 0 \quad (4)$$

$\nabla h(x^k)^T \nabla h(x^k)$ is invertible

$$\nabla f(x^*) + \lambda^* \nabla h(x^*) = 0$$

multiply $\nabla h(x^k)^T$ on both sides of (4)

$$\nabla h(x^k)^T \nabla f(x^k) + 2k (\nabla h(x^k)^T \nabla h(x^k)) h(x^k) + 2 \nabla h(x^k)^T (x^k - x^*) = 0$$

$$\Rightarrow (\nabla h(x^k)^T \nabla h(x^k))^{-1} \nabla h(x^k)^T \nabla f(x^k) + 2k h(x^k) + 2 (\nabla h(x^k)^T \nabla h(x^k))^{-1} \nabla h(x^k)^T (x^k - x^*) = 0$$

letting $x^k \rightarrow x^*$

$$(\nabla h(x^*)^T \nabla h(x^*))^{-1} \nabla h(x^*)^T \nabla f(x^*) + \lim_{k \rightarrow \infty} 2k h(x^k) = 0$$

$$\lim_{k \rightarrow \infty} 2k h(x^k) = \lambda^*$$

$$(\nabla h(x^*)^T \cdot \nabla h(x^*))^T \nabla h(x^*) \cdot \nabla f(x^*) = -\lambda^*$$

$$\nabla f(x^k) + 2k \cdot \nabla h(x^k) \cdot h(x^k) + 2(x^k - x^*) = 0$$

Also pass limit let $x^k \rightarrow x^*$

$$\nabla f(x^*) + \lim_{k \rightarrow \infty} 2 \cdot k \cdot \underline{\nabla h(x^k)} \cdot h(x^k) = 0$$

$$\nabla f(x^*) + \underbrace{\lim_{k \rightarrow \infty} 2 \cdot k \cdot h(x^k)}_{\lambda^*} \cdot \nabla h(x^*) = 0$$

$$\Rightarrow \nabla f(x^*) + \lambda^* \nabla h(x^*) = 0$$

Then let's show @ 8 (b)

$$f^k(x) = f(x) + k|h(x)|^2 + \frac{\alpha}{2} \|x - x^*\|^2$$

$$x^k = \arg \min f^k(x) \quad \text{s.t.} \quad \|x - x^*\|^2 \leq \delta$$

Question: $k \rightarrow \infty$ What will happen:

Fact 1: x^k is a global min.

x^* is a feasible solution

$$\begin{aligned} &\rightarrow f(x^k) + k|h(x^k)|^2 + \frac{\alpha}{2} \|x^k - x^*\|^2 \\ &\leq f(x^*) + k|h(x^*)|^2 + \frac{\alpha}{2} \|x^* - x^*\|^2 \\ &= f(x^*) \quad \forall k \end{aligned}$$

$$x^* \cdot x^*$$

finite number

$$\Rightarrow \lim_{k \rightarrow \infty} |h(x^k)| \rightarrow 0$$

Fact 2:

let $\bar{x} = \lim_{k \rightarrow \infty} x^k$ such a limit must exist

because x^k always satisfy $\|x^k - x^*\| \leq \delta$

$h(\bar{x}) = 0$ it is a feasible solution to the original problem $\min f(x) \text{ s.t. } h(x) = 0$

x^* is local min in radius δ .

$$\left. \begin{aligned} f(x^*) &= f(\bar{x}) \\ x^* &= \bar{x} \end{aligned} \right\} \Rightarrow$$

$$\rightarrow f(x^*) \leq f(\bar{x})$$

$$f(\bar{x}) + \lim_{k \rightarrow \infty} k|h(x^k)|^2 + \frac{\alpha}{2} \|\bar{x} - x^*\|^2 \leq f(x^*)$$

≥ 0